

Database Credit Final 2025 - OCR Extracted Text

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In distributed datafiases. if ey ery node in the cluster can perform the same set of tasks ‘

>>) Hom g€n€Ous Nodes _b) Heterogeneous Nodes) both are correct core 43.

; : ics cated, or

Meaning that users should not be required to know where data is physically locate how

29- tables are partitioned .

a) data consistency b) data integrity c) data transparency d)homogenity of data 44- a)

Each node stores one table, assuming enough storage space for a given node. a

30- ! «2a SOrreeeee

a)horizontal partitioning —b)naive data partitioning __c)vertical Partitioning _d) no correct answer WI

This problem occurs when one transaction updates a database item and then the transaction fajjs 45-

- | The a) It

Ji- ne lcmporan c . ncomrec lnet

oe att at b) The Lost Updaie ais a Unrepeatable :

Updat summary Fad 46. It is

"S1: r2(X) and w1(X): \$2: rl QV and 12(X): which of the two transactions cause conflict

32- a

) S\$] b) 82 c) both d) none 47- The s

A schedule is said to be serializable if it has in the precedence graph a) hiet

Whict

a) cycles b) no cycles oe oF Mae rae cycles :

cycle a) §

at i, ee id 48- WHER

Which ACID property ensures that a transaction brings the database from one valid state to wet fa Sei it

34. another? th an gri

1an gre

a) Atomicity b) consistency €) isolation d) durability Which

All the operations in the transaction are completed successfully and their effect is recorded 49- CREA

arc =

35- permanently in the database

5 ; . What is 1

a) adored b) committed c) denied d) failed < “

S()-

Transaction processes are concurrently executed in multiple CPUs. 2) to crea

36- i re d - :

8) Parallel Processing + b) Interleaved Processing c) common processing d) delayed processing

Q2- Answer th

Which problem does a distributed DBMS not aim to solve?

a) Single point of failure b) Data redundancy c) Geographic latency d) Scalability limits

What is true about shared-disk Systems? Customers

8 a) Nodes share memory b) each node has private c) Data is never d) Used only in Customer! memory persisted parallel DBMS

39. In any data reshuffling in Partitioning when nodes are added/removed we use Orders

4) Consistent hashing b) logical partitioning c) physical Partitioning d) deadlock | OrderID | Ci

Which ACID Property ensures that transactions do not interfere with each other?

a) Atomicity b) consistency c) isolation d) durability OrderItems

j What is the first step in ER-to-Relational mapping? | | Item/2 / Orde

" 8) Mapping of Weak p) Mapping of c) Mapping on i .

a , and i Mapping of Binary d) Mapping of -

Entity Types Multi ? Attributes 4 Relationship Regular Entity Types A) Create the te
ip by add ' hk

If we map a relationship by adding a foreign to the same table referencing its primary key, m

42. we are mapping é B) Count how

= : > ratios

a) 1:1 relationship b) 1:M relationship c) M:N relationship d) Recursive

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)) Domain of

= d) Schema

It is the intermediate layer between the database and the Programs that

14. be Access the data.

;) Metadata by DBMS c) View .

d) Schema

Which of the following is not a function of DBMS

15- a) Data storage b) Data c) Data analysis and management manipulation statistics d) Security management

The ANSI-SPARC architecture divides a database into how many levels?

16- c) 3

a) 3 b) 4 c) 5 d) 6

Which of the following is a key characteristic of a distributed database?

17- j : :

a) Centralized control b) single-point failure c) a network splitting and connectivity required replication

In a 1:1 relationship mapping where one side is optional and the other is mandatory, how many tables are created?

18- a) 1 b) 2 c) 3 d) 4

In a 1:N relationship mapping where the N-side is mandatory, where is the foreign key placed?
9.

19- a) In the table on the 1- b) In the table on the N- by. side See c) In a separate table d) Nowhere

Which of the following is NOT a step in ER-to-Relational mapping?

' - o — .

20- a) Mapping of Regular b) Mapping of Ternary c) Mapping of Unary d) Mapping of Entity Types Relationship Types Relationship Derived Attributes

If XY in a relation, what does this imply?

21- a) X and Y are b) Values of Y are c) Y must be a prime

a) Y determines X b) X determines Y

composite keys determined by X attribute

What is removed to achieve 3NF?

transitive dependencies

22- Functional ;

transitive dependency c) All composite keys d) foreign keys

dependency} . =

A prime attribute is one that:

23- enend: 'oreign b Is part of any c) . Contains NU

a) Depends on a foreign) Sp ontains NULL '7: 254

key candidate key values. d) Violates 2NF

A normal form ensures that every non-prime attribute is fully functional dependent on the whole

24. primary key

a) 1NF 5) 2NF c) 3NF d) BCNF

By normalizing relation or sets Of relations, onesinimizes

25 2) Consistency b) redundancy c) fields d) integrity

A functional dependency is a relationship between or among:

c

26- .) gables b) rows c) relations d) attributes

A relation is 1NF if it doesn't contain any cea

ac in the d)functional

27- inant c)null ~ values * fe ;

- a) peterminants b) repeating groups primary key dependencies

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* M Database systems

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ai of Computer & Infermation Sciences Sr 75/6/2025, Time: 2 hours

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4 € epartment. ; Total Marks: 60 marks

—. o. of pages: 4 pages Model 1 Rath

Answer All questions (Total Marks: 60).

Q1- Choose the correct answer [50 Marks]

Che number of tuples of the relation is called) s,s

a) Degree b) Cardinality c) domain d) Rows

Which of the following is a disadvantage of file-based systems?

* : J

* (5) Data sharin is efficient b) Data redundancy is c) Separation and d) Automatic

= a : minimized, isolation of data concurrency control

. VAn__ integrity requires the foreign key value to match the primary key value or be null.

a) Primary : b) Referential c) Entity d) Special

If a member of a superclass can be a member of one or more subclasses, we use the letter __

4. _ in the circle to describe this relationship.

a) (0) b) (d) c)(t) e&) (m)

Which characteristic of the database approach ensures that changes to the data structure do not

q require changes to application programs?

>

b) Program-dats Date isolatic 7 lti ;

a) Data redundancy BF ened Lately "pi se solation +) \multiple views

pet cence

The process of minimizing the differences between members of an entity by identifying their

6- common characteristics is _ 4

a) Specialization, b) Mining c) Normalization d) Generalization

Metadata in a database system refers to:

7- : b) Data about the c) Backup copies of the

a) The actual stored data co és i Sd) User queri

¥ structure of the data database 0) Vet

In the database environment, who is responsible for giving access and monitoring database use?

8- b Database c

b) atabase cc) Application

a) System analyst m : 1) iy

sig Bane Administrator (DBA) programmer d) End user

Which of the following is NOT a component of a database system?

a) Application programs b) DBMS software c) Metadata d) Manual _ filing
cahinets

= The attribute AGE is calculated from DATE_OF_BIRTH. The age attribute is :

a) multi-value b) Single-value c) Composite d) Derived

rid The partial participation by entities is represented in the E-R diagram as _

c > line " ste line y H 7 CF

a) Single line b) Double line C) Dashed line d) Circle

12 Which architecture allows each node to have its own CPU, memory, and disk?

a) Shared Everything b) Shared Memory c) Shared Disk d) Shared Nothing

13- For each attribute of a relation, there is a set of permitted values, called the e ;

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Page 4

relationship

Q5. An entity set that includes one or more distinct, subgroupings of its occurrences or how b) superclass c) child class d) member class

A specialization lattice differs from a hierarchy because:

44 a) multiple inheritance prohibits c) it only supports d) it is not

superclasses allow single level subclasses constraints &

answer Which of the following is true about a leaf node in a Specialization hierarchy?

45- b) It must have at least one child

a) It has no subclasses b) It must be a child c) It cannot inherit d) It is always

superclasses attributes optional

46- a) It is possible for a member of a superclass to be a member of only one subclass,

a) specialization b) generalization c) overlapping d) disjoint

The specialization constraint that each subclass has only one parent

47-

a) hierarchy b) lattice c) union d) leaf

Which of the following is true about the HAVING clause?

48- a) Similar to the WHERE clause b) Similar to WHERE ;

c) Similar to WHERE ; d) Similar to WHERE

49- WHERE clause but is used for rows. HAVING clause is used for groups. HAVING clause is used for groups.

State to which clause is used for

50- a) WHERE clause b) HAVING clause c) GROUP BY clause d) ORDER BY clause

used for columns rather than rows rather than groups rather than rows. TERNARY

than groups. columns

51- Which command is used to change the definition of a table in SQL?

» 49- :

recorded a) CREATE b) UPDATE c) ALTER d) SELECT

What is the primary purpose of the GRANT command in SQL?

50- a) to delete data from a table b) to give user access d) to update table

a) to create new table b) table privileges data

Q2- Answer the following questions [10 Marks]

Centralized Based on the following tables solve the following questions:

limits

Customers

only in | CustomerID | Name | City |

BMS

Orders ;

o stomerID | OrderDate |

x | OrderID | Cu

rability OrderItem 5

smip | Orderib | ProductName | Quantity | :

| || itemi2 |} ===

a ”

pping of the table “Customers”

antity Types A Create

” ”

rimary key, t how man orders "Sara" has made.

B) Coun

ecursive

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